

Question				Marking details	Marks available																																							
					AO1	AO2	AO3	Total	Maths	Prac																																		
6	(a)			beaker ☐ thermometer ☐ gas syringe ☒ conical flask ☐	1			1		1																																		
	(b)	(i)		<table><tr><th rowspan="2">Concentration of acid (M)</th><th colspan="4">Volume of gas produced (cm³)</th></tr><tr><th>Test 1</th><th>Test 2</th><th>Test 3</th><th>Mean</th></tr><tr><td>0.2</td><td>16</td><td>14</td><td>15</td><td>15</td></tr><tr><td>0.4</td><td>31</td><td>33</td><td>30</td><td>32</td></tr><tr><td>0.6</td><td>47</td><td>49</td><td>29</td><td>48</td></tr><tr><td>0.8</td><td>63</td><td>64</td><td>65</td><td>64</td></tr><tr><td>1.0</td><td>82</td><td>83</td><td>79</td><td>81</td></tr></table> award (1) for identifying 29 award (1) for reason e.g. it is too far away from the others (at that concentration) it is an anomaly it is too far from the mean it is too low / it is much lower than the others neutral answers – it is lower (than the others) / unreliable / incorrect	Concentration of acid (M)	Volume of gas produced (cm ³)				Test 1	Test 2	Test 3	Mean	0.2	16	14	15	15	0.4	31	33	30	32	0.6	47	49	29	48	0.8	63	64	65	64	1.0	82	83	79	81			2	2		2
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		(ii)		award (2) for all five points plotted correctly (± 1 square) award (1) for any three correct points award (1) for straight line		2	1	3	3	3
		(iii)		40 accept any value in the range 39-41 ecf possible from incorrect line (± 1 square)		1		1	1	
	(c)			there are more particles of acid ☐ the acid particles are moving faster ☒ there is a higher surface area ☐ there are more collisions per second ☒ the acid particles have less energy ☐ award (1) for each correct answer deduct (1) for each additional tick if more than two ticks		2		2		2
	(d)			hydrogen accept H ₂	1			1		1
				Question 6 total	2	5	3	10	4	9